



RESEARCH PAPER



SkillClaw

Collective Skill Evolution for
Multi-User LLM Agent Ecosystems



Let Skills Evolve Collectively with Agentic Evolver



Agent Skills Remain Static After Deployment

Users rediscover solutions independently



Current State: Static

Skills never improve after deployment

- Skills manually installed from hub
- Never updated from runtime experience
- Failures rediscovered repeatedly
- Solutions lost after individual sessions
- No system-level knowledge accumulation

Memory-based: tied to instances, hard to generalize

Skill-based: library treated as static resource



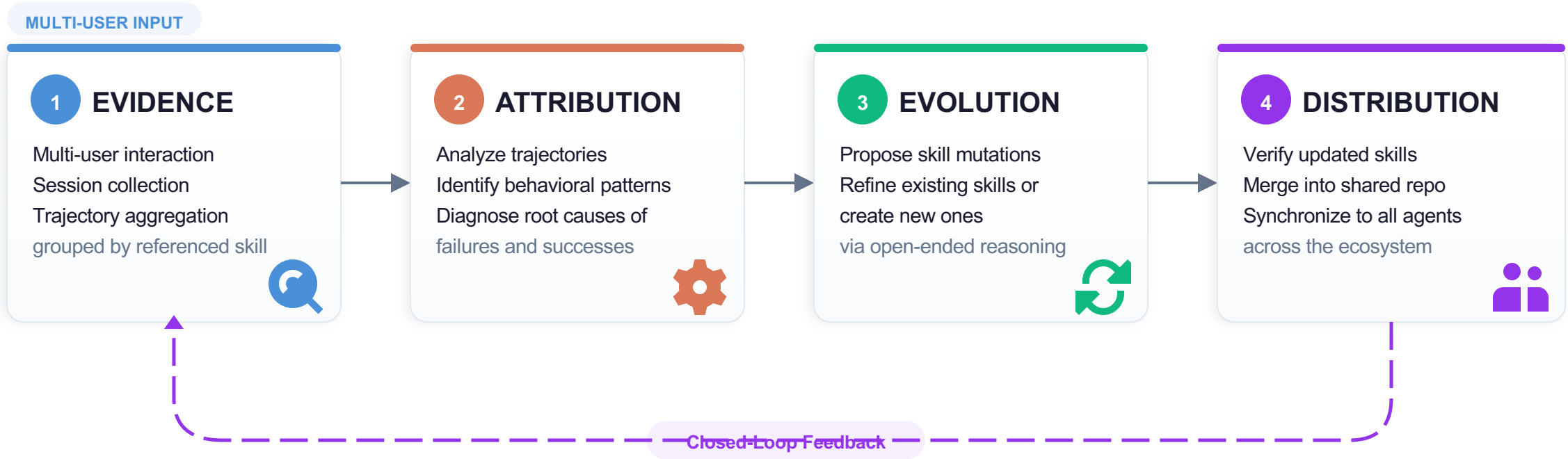
Needed: Evolving

Skills should improve from collective usage

- Skills improve from runtime interaction
- Learnings shared across all users
- Solutions persist and propagate
- Continuous ecosystem improvement
- Knowledge accumulates at system level

SkillClaw: collective evolution + automatic +
agentic reasoning over interaction evidence

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution

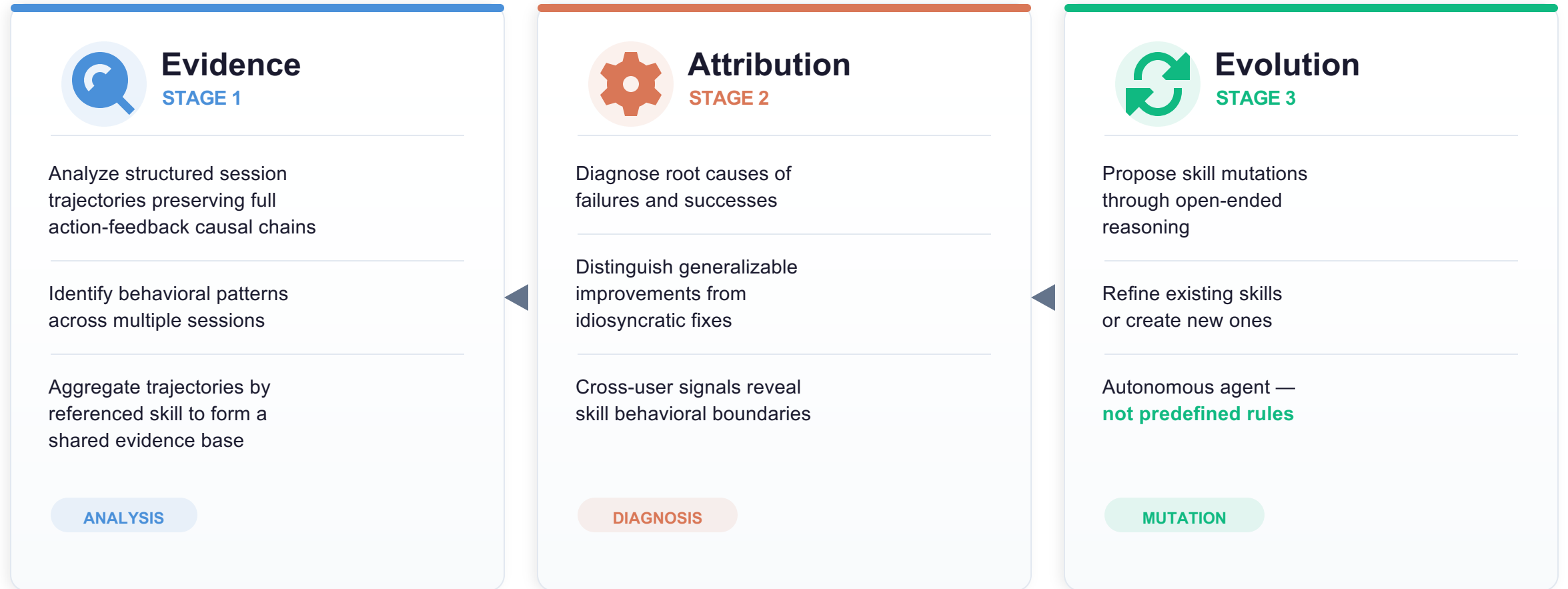


Key Insight

Evolution is driven by an autonomous agent performing open-ended reasoning over interaction evidence — not predefined update rules. The closed loop ensures continuous improvement.

Agentic Evolver: Evidence → Attribution → Evolution

Three-stage open-ended reasoning, not predefined rules



Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

1 Complementary Behavioral Signals

Different users exercising the same skill under diverse contexts produce complementary views of that skill's behavioral boundary.

2 What Aggregation Reveals

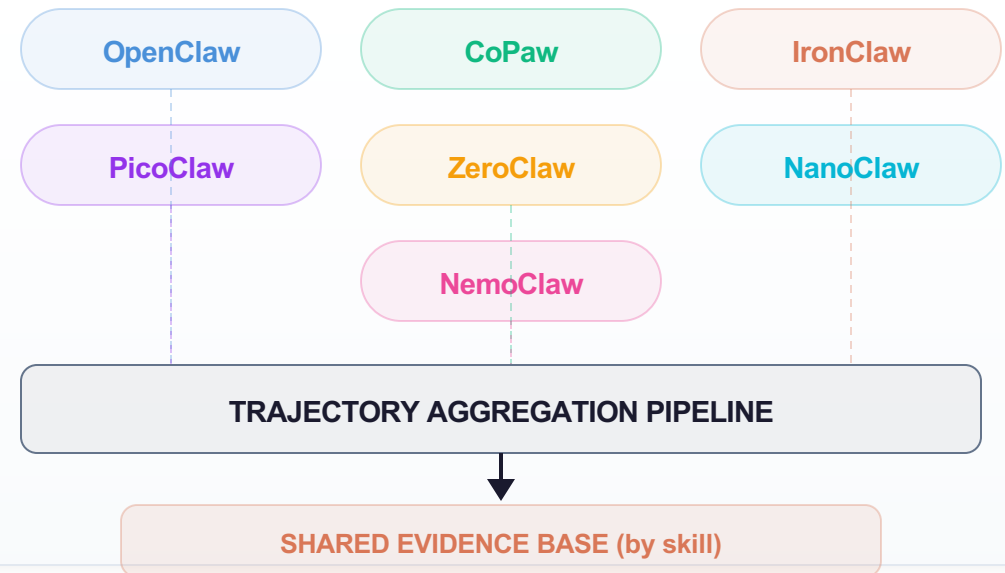
- Conditions where the skill works
- Conditions where the skill breaks

3 Single-User Limitation

A single user rarely generates enough signal to separate generalizable improvement from

Compatible Claw-Style Agent Systems

All feed into shared trajectory aggregation



Key Finding

Complementary behavioral signals from diverse users enable robust skill evolution that no single user could achieve alone

Multimodal Creative Pipeline Skill

6 Days of Evolution Attempts — Only 1 of 6 Accepted

Day	Candidate Skill	Skill Function / Change	Validator	Next-Day Action
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & environment setup	ACCEPT	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	REJECT	Keep current best pool
3	multimodal-creative-task-pipeline	Multimodal creative pipeline (extract from PDF/video/image)	REJECT	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Added image classification, visual gen., structured validation	REJECT	Keep current best pool
5	creative-pipeline (impr.) + validate-required-input-files	Creative pipeline + per-file fail-fast validation	REJECT	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / document-to-visual paths	REJECT	Not admitted to next cycle

1 of 6 Accepted

— Conservative verification prevents regressions. Only Day 1's workspace validation passed; five subsequent candidates were rejected despite increasingly complex proposals.

Git Push Auth-Fallback Skill

Iterative Refinement Over 6 Days — 3 of 6 Accepted

Day	Candidate Skill	Change Summary	Validator	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	ACCEPT	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	ACCEPT	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	ACCEPT	Keep current best pool
4	(none — same-pool retest)	Same-pool retest; validator confirmed current pool as best	ACCEPT	Continue same best pool
5	git-push-with-auth-fallback	Added "push hang treated as auth failure"; no improvement	REJECT	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	REJECT	Not admitted to next cycle

3 of 6 Days Accepted

— Successful iterative refinement before plateauing. Days 1–3 progressively improved the skill; Days 5–6 changes failed to improve over the established best pool.

Three Properties That Distinguish SkillClaw

From existing memory-based and skill-based systems



Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem.

vs. memory-based: tied to specific instances, hard to generalize across users

SHARED

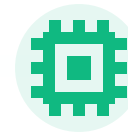


Fully Automatic

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention.

vs. skill-based: library treated as static resource, requires manual updates

AUTONOMOUS



Agentic Evolution

Skill updates are produced through open-ended reasoning rather than predefined update rules.

Enables context-aware skill mutations that adapt to emerging failure patterns

REASONING

Implication

Build agent systems that continuously improve from collective usage — not just individual sessions.